CLASSIFYING GROTHENDIECK RINGS OF PIVOTAL FUSION CATEGORIES OF RANK FOUR

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Abstract. We classify the Grothendieck rings of pivotal fusion categories of rank four over the complex field. There are exactly fifteen, each admitting a unitary categorification. This classification is obtained in two stages. The first consists in proving that the global Frobenius–Perron dimension of the Grothendieck ring of any pivotal fusion category of rank four is less than 3600. The second uses known and new exclusion criteria to show that, among all fusion rings of rank four with global Frobenius–Perron dimension less than 3600, only fifteen admit a pivotal categorification.

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1. Introduction

Fusion categories over the complex field connect representation theory and subfactor theory with low-dimensional topology and mathematical physics. Spherical fusion categories furnish state-sum invariants of three-manifolds [2] and unitary fusion categories provide the algebraic input for string-net models of topological phases [18]. Their classification has a strong arithmetic aspect: dimensions and formal codegrees are algebraic integers constrained by Galois theory and divisibility [9, 25]. We refer to [8] for the foundations and standard conventions of fusion category theory.

The *rank* of a fusion category counts the isomorphism classes of its simple objects; its *multiplicity* is the largest coefficient in the decomposition of a tensor product of two simple objects. Fixing the rank alone does not bound the multiplicity of an abstract fusion ring: the rank-two rings with $x^2 = 1 + ax$ exist for every $a \in \mathbb{Z}_{\geq 0}$. A finite classification of Grothendieck rings of fixed rank therefore requires constraints coming from categorifiability. In rank four, we obtain, thanks to the interaction between arithmetic constraints, the characters of Grothendieck rings, and the

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Frobenius-Schur indicators, that the global Frobenius–Perron dimension of the Grothendieck ring of any pivotal fusion category of rank four is less than 3600. It leads to the following classification.

Theorem 1.1. *Up to fusion ring isomorphism, there are fifteen fusion rings of rank four that admit a pivotal categorification; they are listed in Table 1. Each of them admits a unitary categorification.*

The theorem classifies fusion rings, with all categorifications of the same fusion ring counted once. It also establishes that the existence of a pivotal categorification is equivalent, in rank four, to the existence of a unitary categorification.

$\mu(R)$	A unitary realization	FPdim($\mathcal{C}$)	Approx.	Reference
1	$\mathrm{Vec}_{C_2\times C_2}$	4	4.000	[20, §3.4]
1	Vec_{C_4}	4	4.000	[20, §3.4]
1	Tambara–Yamagami on C_3	6	6.000	[30]
1	$\mathrm{Fib}\boxtimes\mathrm{Vec}_{C_2}$	$5+\sqrt{5}$	7.236	[20, §3.4]
1	$\mathrm{Rep}(D_5)$	10	10.000	[20, §3.4]
1	$\mathrm{Fib}\boxtimes\mathrm{Fib}$	$(15+5\sqrt{5})/2$	13.090	[20, §3.4]
1	$\mathrm{PSU}(2)_6$	$8+4\sqrt{2}$	13.657	[7, Theorem 3.1]
1	Principal even part of S'	$8+4\sqrt{2}$	13.657	[19, Theorem 1]
1	$\mathrm{PSU}(2)_7$	δ_7	19.234	[20, §3.4]
2	$\mathrm{Rep}(A_4)$	12	12.000	[17, Theorem 1.1]
2	$\mathcal{H}_1$ (Haagerup)	$(39+9\sqrt{13})/2$	35.725	[11, Table 2]
2	Principal even part of $2D2$	$20+8\sqrt{5}$	37.889	[7, Theorem 3.1]
3	Near-group type $(C_3, 3)$	$(21+3\sqrt{21})/2$	17.374	[16, Example 9.4]
3	$\mathrm{Ad}(E_8)$	δ_E	35.238	[15, 10]
6	Near-group type $(C_3, 6)$	$24+12\sqrt{3}$	44.785	[16, Theorem 10.19]

TABLE 1. One unitary realization of each surviving fusion ring, with its multiplicity and global Frobenius–Perron dimension (rounded to three decimals in the fourth column). Rows are ordered first by multiplicity and then by increasing global Frobenius–Perron dimension. The numbers δ_7 and δ_E are the largest real roots of $x^3-27x^2+162x-243$ and $x^4-45x^3+375x^2-1125x+1125$, respectively. The table also records references for the construction of each of these fifteen fusion rings.

Remark 1.2. For pivotal fusion categories of rank four, Table 1 gives the sharp bounds $\mu(K_0(\mathcal{C}))\leq 6$ and $\mathrm{FPdim}(\mathcal{C})\leq 24+12\sqrt{3}$, both attained by the near-group fusion ring of type $(C_3, 6)$.

From now on, all fusion categories are over $\mathbb{C}$. Ostrik classified fusion categories of rank two [24] and pivotal fusion categories of rank three [26]. In rank four, the modular and premodular cases were studied in [28, 13, 3]. Without braiding, Larson [17] reduced the non-self-dual pseudo-unitary case to seven fusion rings. Dong–Zhang–Dai [6] treated the nontrivially graded self-dual spherical case, and Edie-Michell–Izumi–Penneys [7] classified $\mathbb{Z}/2\mathbb{Z}$-quadratic unitary fusion categories. In particular, the ring $S(2)$, already present as $K_2(2)$ in Larson's list [17, Theorem 1.1(6)], was excluded from unitary categorification. Example 3.4 explains how a short calculation allows to rule out $S(2)$ from a pivotal categorification. Further classification results appear in [5, 4].

We now conclude this introduction by outlining the organization of the paper. Section 2 sets out the basic terminology for fusion rings and fusion categories, then recalls several well-known results on categorifiable fusion rings. Section 3 establishes new inequalities, based on the Frobenius–Schur indicators, that must be satisfied by any pseudo-unitary fusion category of a given rank. Section 4 contains the crucial statement: the global Frobenius–Perron dimension of any pivotal fusion category of rank four is less than 3600. The non-self-dual case follows from Larson [17], the self-dual pivotal, but not pseudo-unitary, case is handled by an arithmetic argument, and the remaining case

involves inequalities coming from Section 3. The final Section 5 provides Python commands to find all fusion rings of rank four whose global Frobenius–Perron dimension is less than 3600; there are 8977. However, with the help of other commands, we will see that 8962 of them violate at least one of the properties of pivotal fusion categories stated in Sections 2 and 3. Theorem 1.1 then follows.

2. Preliminaries on fusion rings and fusion categories

2.1. **Fusion rings and formal codegrees.** A *fusion ring* R is a free $\mathbb{Z}$-module of finite rank with basis $\{b_0 = 1, b_1, \dots, b_{r-1}\}$ and multiplication

$$b_i b_j = \sum_k N_{ij}^k b_k,$$

where $N_{ij}^k \geq 0$ are integers, called the *fusion coefficients*, satisfying:

- (Associativity) $\forall i, j, k$, one has $b_i(b_j b_k) = (b_i b_j) b_k$;
- (Unit) $\forall i$, one has $b_0 b_i = b_i b_0 = b_i$;
- (Dual) $\forall i, \exists! i^* \in \{0, \dots, r-1\}, \forall j$ one has $N_{ij}^0 = N_{ji}^0 = \delta_{i^* j}$, where δ_{lm} is the Kronecker symbol;
- (Frobenius reciprocity) $\forall i, j, k$, one has $N_{ij}^k = N_{i^* k}^j = N_{kj^*}^i$.

Put $N_i := (N_{ij}^k)_{j,k}$. Then $N_i^{\mathsf{T}} = N_{i^*}$. We say that R is *self-dual* if $i^* = i$ for all i. The *Frobenius–Perron dimension* is the unique ring homomorphism $\mathrm{FPdim} : R \to \mathbb{R}$ that is positive on the distinguished basis. The spectral radius of N_i is $d_i := \mathrm{FPdim}(b_i)$ [9, Theorem 8.2 and Lemma 8.3]. Write

$$\mathrm{FPdim}(R) = \sum_i d_i^2 \qquad \text{and} \qquad \mu(R) = \max_{i,j,k} N_{ij}^k.$$

A fusion ring of rank at most five is commutative by [1, Proposition 9.2]. Thus, what follows is valid for $r = 4$ or $r = 5$.

Let R be a commutative fusion ring of rank r. It then has r characters, denoted by $\chi_0, \dots, \chi_{r-1}$, and one of them is the homomorphism FPdim, say χ_0 [8, Proposition 3.3.6 (3)]. Given j, the *formal codegree* of χ_j is the real number $f_j = \sum_{i=0}^{r-1} |\chi_j(b_i)|^2$. Since $\chi_j(b_i)$ is an eigenvalue of N_i for every i, we infer that f_j is an algebraic integer and that $|\chi_j(b_i)| \leq d_i$; whence $f_j \leq \mathrm{FPdim}(R) = f_0$. Also, χ_j is a real character if R is self-dual, since N_i is then a real symmetric matrix for every i.

The normalized character vectors $(\chi_j(b_i))_i / \sqrt{f_j}$ form an orthonormal basis, giving

$$\sum_{j=0}^{r-1} \frac{\chi_j(b_i)\overline{\chi_j(b_k)}}{f_j} = \delta_{ik} \qquad \text{and} \qquad \sum_{j=0}^{r-1} \frac{1}{f_j} = 1. \tag{2.1}$$

See [21, Theorem 2.4] for the first equality. The second is obtained from the first one by taking $i = k = 0$. In particular, $f_j > 1$ if $r \geq 2$.

Lemma 2.1. *Let R be a commutative fusion ring of rank r. Its fusion matrices are normal and the eigenvalues of its* Casimir matrix $\sum_i N_i N_i^{\mathsf{T}}$ *are the formal codegrees $f_0, \dots, f_{r-1}$. In addition, $\mu(R)^2 \leq \mathrm{FPdim}(R)$.*

Proof. The Casimir spectrum is given by [25, Example 2.4 and Lemma 2.6]. The normality of fusion matrices arises from their pairwise commutativity and from $N_i^{\mathsf{T}} = N_{i^*}$. Hence, the Euclidean operator norm of N_i equals the spectral radius of N_i, namely d_i, and so $N_{ij}^k \leq d_i \leq \sqrt{\mathrm{FPdim}(R)}$. □

2.2. **Generalities on fusion categories.** For a fusion category $\mathcal{C}$, let $\mathcal{O}(\mathcal{C})$ be a set of representatives of its simple isomorphism classes and write $V_0 = \mathbf{1}, \dots, V_{r-1}$ for these representatives. Its Grothendieck ring $K_0(\mathcal{C})$ is a fusion ring with distinguished basis $\{1, [V_1], \dots, [V_{r-1}]\}$ and with multiplication induced by $\otimes$ [8, Proposition 4.9.1]. We set $\mathrm{FPdim}(\mathcal{C}) = \mathrm{FPdim}(K_0(\mathcal{C}))$. Its *categorical*

dimension is

$$\dim(\mathcal{C}) = \sum_{V \in \mathcal{O}(\mathcal{C})} |V|^2,$$

where $|V|^2$ is the squared norm of V [8, Definition 7.21.3]. We say that $\mathcal{C}$ is *pseudo-unitary* if $\dim(\mathcal{C}) = \mathrm{FPdim}(\mathcal{C})$. It is known that $\mathcal{C}$ is pseudo-unitary if $\mathrm{FPdim}(\mathcal{C}) \in \mathbb{Z}$ [8, Proposition 9.6.5].

Applying any field automorphism σ of $\mathbb{C}$ to the structure morphisms gives the Galois-conjugate fusion category $\mathcal{C}^\sigma$. The fusion coefficients and the set of simple objects remain unchanged, so $K_0(\mathcal{C}^\sigma) \cong K_0(\mathcal{C})$ and $\mathrm{FPdim}(\mathcal{C}^\sigma) = \mathrm{FPdim}(\mathcal{C})$. One also has $\dim(\mathcal{C}^\sigma) = \sigma(\dim(\mathcal{C}))$ [27, §2.1].

A *pivotal structure* on $\mathcal{C}$ is a monoidal natural isomorphism $j : \mathrm{Id}_{\mathcal{C}} \Rightarrow (-)^{**}$ [8, Definition 4.7.7]. It defines a dimension character χ on $K_0(\mathcal{C})$ by $\dim_{\mathcal{C}}(V) := \chi([V])$, which is nonzero on every simple object V [8, Propositions 4.7.12 and 4.8.4]. Moreover,

$$\dim(\mathcal{C}) = \sum_{V \in \mathcal{O}(\mathcal{C})} \dim_{\mathcal{C}}(V) \dim_{\mathcal{C}}(V^*).$$

The pivotal structure is said to be *spherical* if $\dim_{\mathcal{C}}(V) = \dim_{\mathcal{C}}(V^*)$ for every simple object V [8, Definition 4.7.14]. For a commutative Grothendieck ring, this is equivalent to these dimensions being real. A pseudo-unitary fusion category admits a canonical spherical structure with simple dimensions equal to $1, \mathrm{FPdim}([V_1]), \ldots, \mathrm{FPdim}([V_{r-1}])$ [8, Proposition 9.5.1].

2.3. **Reductions.** Theorem 4.1 establishes that any fusion ring R of rank four admitting a pivotal categorification must satisfy $\mathrm{FPdim}(R) < 3600$. To obtain such a bound, we need to reduce the problem to a setting with more favorable hypotheses. This is the purpose of this subsection. For example, for rank-four fusion rings, pivotal categorifiability and spherical categorifiability coincide.

Lemma 2.2. *Every pivotal fusion category of rank four admits a spherical structure.*

Proof. A pivotal fusion category admitting no spherical structure has at least five isomorphism classes of simple objects [12, §7, Lemma 4]. □

We now explain that if $\dim(\mathcal{C})$ and $\mathrm{FPdim}(\mathcal{C})$ have the same minimal polynomial over $\mathbb{Q}$, then we may assume without loss of generality that $\mathcal{C}$ is pseudo-unitary.

Lemma 2.3. *Let $\mathcal{C}$ be a fusion category. If $\dim(\mathcal{C})$ and $\mathrm{FPdim}(\mathcal{C})$ are Galois conjugate, then $\mathcal{C}$ has a pseudo-unitary Galois conjugate.*

Proof. Choose a field automorphism σ of $\mathbb{C}$ with $\sigma(\dim(\mathcal{C})) = \mathrm{FPdim}(\mathcal{C})$. The preceding identities give $\dim(\mathcal{C}^\sigma) = \sigma(\dim(\mathcal{C})) = \mathrm{FPdim}(\mathcal{C}) = \mathrm{FPdim}(\mathcal{C}^\sigma)$. □

2.4. **Restrictions.** As we will see in Section 5, there are 8977 fusion rings of rank four whose global Frobenius–Perron dimension is less than 3600. We can now check one by one whether these fusion rings admit a pivotal categorification. This subsection reviews exclusion criteria that reduce the list of potential candidates to 75. The indicator inequalities of the next section will rule out another 60.

Theorem 2.4. (Ostrik, [24, Main Theorem, p. 178]) *A fusion ring with basis $\{1, x\}$ and fusion rule $x^2 = 1 + ax$ is categorifiable only for $a = 0, 1$.*

A nonzero algebraic integer α is called a *unit* if α^{-1} is an algebraic integer and a *d-number* if β/α is a unit for all Galois conjugates β of α. Letting n be the degree of α over $\mathbb{Q}$, the element α is a d-number if and only if $\alpha^n = k\varepsilon$ for some nonzero integer k and some unit ε [25, Lemma 2.7(iv)].

Theorem 2.5. (Ostrik, [25, Theorem 1.2]) *Every formal codegree of a categorifiable fusion ring is a d-number.*

We also use three published rank-four classifications. In each nonpointed family below, the basic Frobenius–Perron dimensions take exactly the values $1, d$, with $d > 1$. Any categorification is therefore Galois conjugate to a pseudo-unitary one [29, Lemma 3.1]. The following are necessary conditions for categorifiability:

(1) For the near-group ring $\mathrm{NG}(C_3, m)$, with basis $C_3 \sqcup \{x\}$ and rules $gx = xg = x$, $x^2 = \sum_{g\in C_3} g + mx$, one has $m \in \{0, 2, 3, 6\}$ [17, Theorem 1.1].
(2) For the ring with basis $\{1, g, x, x^*\}$ and rules $g^2 = 1$, $gx = x^*$, $x^2 = g + m(x + x^*)$, one has $m \leq 2$ [17, Theorem 1.1].
(3) For the self-dual ring with basis $\{1, g, x, y\}$ and rules $g^2 = 1$, $gx = y$, $x^2 = 1 + ax + by$, one has $(a, b) \in \{(0, 0), (0, 1), (1, 0), (1, 1), (2, 2)\}$ [7, Theorem 2.4].

Up to fusion ring isomorphism, Theorem 1.1 classifies all fusion rings of rank four admitting a pivotal categorification. Any such categorification must be spherical by Lemma 2.2. Hence, for our purposes, it is particularly relevant to study the properties of a spherical fusion category $\mathcal{C}$ whose Grothendieck ring is commutative, as we do in the remainder of this subsection.

Put $c_i = \dim_{\mathcal{C}}(V_i)$ for every $i \in \{0, \ldots, r-1\}$. Note that $c_i = c_{i^*}$ for all i and that $\dim(\mathcal{C}) = \sum_i c_i^2$. The c_i are nonzero totally real algebraic integers [26, §2.1]. Ostrik's spherical inequality yields

$$\sum_j \frac{1}{f_j^2} \leq \frac{1}{2} + \frac{1}{2\sigma(\dim(\mathcal{C}))} \tag{2.2}$$

for all field automorphisms σ of $\mathbb{C}$ [27, Theorem 3.1.1 and Remark 3.1.2]. Also, $\dim(\mathcal{C})/f$ must be a cyclotomic integer for every formal codegree f of $K_0(\mathcal{C})$ [26, Theorem 2.13]. In particular, the norm of $\dim(\mathcal{C})/f$ is an integer and the field generated by $\dim(\mathcal{C})/f$ is abelian over $\mathbb{Q}$. For a ratio of formal codegrees, these conditions also suffice for cyclotomic integrality, as the following lemma shows.

Lemma 2.6. *Let u, v be d-numbers and put $x = u/v$. Then x is a cyclotomic integer if and only if $N_{\mathbb{Q}(x)/\mathbb{Q}}(x) \in \mathbb{Z}$ and $\mathbb{Q}(x)/\mathbb{Q}$ is abelian.*

Proof. The forward implication is clear. Conversely, extend each embedding of $\mathbb{Q}(x)$ to an automorphism σ of $\mathbb{C}$. Since u, v are d-numbers, $\sigma(x)/x = (\sigma(u)/u)(v/\sigma(v))$ is an algebraic-integer unit. Thus $N_{\mathbb{Q}(x)/\mathbb{Q}}(x)$ is $x^{[\mathbb{Q}(x):\mathbb{Q}]}$ times a unit. Its integrality implies that x is an algebraic integer. The abelianity assumption and the Kronecker–Weber theorem place x in a cyclotomic field. □

The abelianity condition can be checked once for the entire ring. Indeed, for any formal codegree H, equation (2.1) gives $H = \sum_j H/f_j$. Hence all fields $\mathbb{Q}(H/f_j)$ are abelian if and only if the splitting field of $\prod_j (t - f_j)$ is abelian. This polynomial is the characteristic polynomial of the Casimir matrix and has degree four in our computation.

3. Central induction and indicator inequalities

Let $\mathcal{C}$ be a pseudo-unitary fusion category, equipped with its canonical spherical structure, and put $D = \mathrm{FPdim}(\mathcal{C})$. Let $F : \mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ be the forgetful functor and I its right adjoint. For a simple object A of the center, write θ_A for its twist and $[I(V) : A]$ for its multiplicity in $I(V)$. We use the following identities:

$$\sum_A [I(V) : A] \,\mathrm{FPdim}(A) = D\,\mathrm{FPdim}(V), \qquad \sum_A [I(V) : A] \,\mathrm{FPdim}(A)\theta_A^n = D\nu_n(V) \quad (n \geq 1).$$

The first follows from $FI(V) \cong \bigoplus_{X\in\mathcal{O}(\mathcal{C})} X \otimes V \otimes X^*$ [26, Proposition 2.4]; the second is the indicator formula [23, Theorem 4.1]. The twists are roots of unity [23, Theorem 5.5], and all simple summands of $I(1)$ have twist one [26, Theorem 2.5].

The *low part* of $I(V)$ consists of its simple summands that also occur in $I(1)$, with their multiplicities in $I(V)$:

$$I_{\mathrm{low}}(V) = \bigoplus_{A\in\mathcal{O}(I(1))} A^{\oplus [I(V):A]}.$$

Here $\mathcal{O}(I(1))$ denotes the simple support of $I(1)$. Write $d_V = \mathrm{FPdim}(V)$ and $W_V = \mathrm{FPdim}(I_{\mathrm{low}}(V))$. For a root of unity ζ, let $I(V)_\zeta$ be the sum of the simple summands of $I(V)$ with twist ζ, again with their multiplicities. Thus $I_{\mathrm{low}}(V)$ is a direct summand of $I(V)_1$.

Proposition 3.1 (Polynomial indicator bounds)**.** *Let $V \in \mathcal{C}$ and let*

$$P(z) = c_0 + \sum_{k=1}^{n} (c_k z^k + \overline{c_k} z^{-k}), \qquad c_0 \in \mathbb{R}, \quad c_k \in \mathbb{C},$$

be nonnegative on the unit circle. Then

$$P(1) W_V \le D \left(c_0 d_V + 2 \operatorname{Re} \sum_{k=1}^{n} c_k \nu_k(V) \right).$$

For every root of unity ζ, one also has

$$P(1) \,\mathrm{FPdim}(I(V)_\zeta) \le D \left(c_0 d_V + 2 \operatorname{Re} \sum_{k=1}^{n} c_k \zeta^{-k} \nu_k(V) \right).$$

Proof. The indicator formula and $|\theta_A| = 1$ give

$$\sum_A [I(V) : A] \,\mathrm{FPdim}(A) P(\theta_A) = D \left(c_0 d_V + 2 \operatorname{Re} \sum_{k=1}^{n} c_k \nu_k(V) \right).$$

Every term on the left is nonnegative. The summands from $I(1)$ all have twist one, so keeping only those terms gives the first inequality. For the second, replace $P(z)$ by $P(\zeta^{-1} z)$ and keep the terms with $\theta_A = \zeta$. □

We obtain the polynomials used below by taking squared moduli. For $Q(z) = \sum_{j=0}^{n} a_j z^j$ and $|z| = 1$,

$$|Q(z)|^2 = c_0 + \sum_{k=1}^{n} (c_k z^k + \overline{c_k} z^{-k}), \qquad c_0 = \sum_{j=0}^{n} |a_j|^2, \quad c_k = \sum_{j=0}^{n-k} a_{j+k} \overline{a_j}.$$

Here $\overline{z} = z^{-1}$; this Laurent expansion is an identity on the unit circle.

Remark 3.2 (Vanishing indicators)**.** If $\nu_1(V) = \cdots = \nu_m(V) = 0$ for an integer $m \ge 1$, apply Proposition 3.1 to

$$|1 + z + \cdots + z^m|^2 = (m+1) + \sum_{k=1}^{m} (m + 1 - k)(z^k + z^{-k})$$

to obtain

$$(m+1) W_V \le D d_V, \qquad (m+1) \,\mathrm{FPdim}(I(V)_\zeta) \le D d_V.$$

For $m = 1$, the low-part bound is the estimate in [26, proof of Theorem 2.21]; the extension to twist sectors was suggested by Andrew Schopieray. For a direct sum of non-self-dual simples, one can take $m = 2$, by additivity of indicators [22, Corollary 7.8].

Corollary 3.3. *For a nonunit simple object X,*

$$9 W_X \le D(3 d_X + 2 \nu_2(X)).$$

In particular,

$$9 W_X \le D(3 d_X + 2) \quad (X \cong X^*), \qquad 3 W_X \le D d_X \quad (X \ncong X^*).$$

Proof. Apply Proposition 3.1 to $|1 + z + z^2|^2 = 3 + 2(z + z^{-1}) + (z^2 + z^{-2})$. Here $\nu_1(X) = 0$, while $\nu_2(X)$ is ± 1 for a self-dual simple and zero otherwise [22, Definition 3.1 and Theorem 5.1]. □

Example 3.4. Corollary 3.3 rules out the ring $S(2)$ of [7, §4, Theorem 4.1], already listed by Larson [17, Theorem 1.1(6)] as $K_2(2)$. It has basis $\{1, \alpha, \rho, \rho^*\}$, with $\alpha^2 = 1$, $\rho^* = \alpha\rho$, and $\rho^2 = \alpha + 2\rho + 2\rho^*$. Put $d = 2 + \sqrt{5}$ and $D = 4 + 8d$; its simple Frobenius–Perron dimensions are $1, 1, d, d$. Any categorification has a pseudo-unitary Galois conjugate [29, Lemma 3.1]. Larson's induction formulas [17, Proposition 5.2], with $c = 2$, then give

$$W_\rho = 2(1 + 4d) + 2(1 + 2d) = 4 + 12d.$$

Since ρ is not self-dual, the corollary would imply

$$12 + 36d = 3W_\rho \le Dd = 8 + 36d,$$

a contradiction. Thus $S(2)$ has no complex fusion categorification.

Corollary 3.5. *For a nonunit simple object X, put $a = \dim_{\mathbb{C}} \mathrm{Hom}(1, X^{\otimes 3}) = N_{XX}^{X^*}$. Then*

$$36W_X \le D(10d_X + 8 + 2a).$$

Proof. Apply Proposition 3.1 to

$$|(1 + z)(1 + z + z^2)|^2 = 10 + 8(z + z^{-1}) + 4(z^2 + z^{-2}) + (z^3 + z^{-3}).$$

Use $\nu_1(X) = 0$, $\nu_2(X) \le 1$ and $|\nu_3(X)| \le a$. The last bound holds because $\nu_n(X)$ is the trace of a rotation on $\mathrm{Hom}(1, X^{\otimes n})$ whose nth power is the identity [22, Definition 3.1 and Theorem 5.1]. □

Corollary 3.6 (Fourth-indicator bound)**.** *For a nonunit simple object X, put*

$$s = \dim_{\mathbb{C}} \mathrm{Hom}(1, X^{\otimes 2}) = \mathbf{1}_{X \cong X^*}, \qquad a = \dim_{\mathbb{C}} \mathrm{Hom}(1, X^{\otimes 3}) = N_{XX}^{X^*}.$$

Then

$$32W_X \le D\left(7d_X + \left\lfloor \frac{10d_X + 71s + 30a}{9} \right\rfloor\right).$$

Proof. The rotation eigenvalues give $\nu_2(X) \in \mathbb{Z}$, $2\,\mathrm{Re}\,\nu_3(X) \in \mathbb{Z}$ and $\mathrm{Re}\,\nu_4(X) \in \mathbb{Z}$. Hence

$$J := 8\nu_2(X) + 4\,\mathrm{Re}\,\nu_3(X) + \mathrm{Re}\,\nu_4(X) \in \mathbb{Z}.$$

Expanding

$$P_+(z) = |(1 + z)^2(1 + z^2)|^2, \qquad P_-(z) = |3 + z - z^3 - 3z^4|^2$$

and applying Proposition 3.1 gives

$$32W_X \le D(7d_X + J), \qquad \nu_2(X) + 6\,\mathrm{Re}\,\nu_3(X) + 9\,\mathrm{Re}\,\nu_4(X) \le 10d_X.$$

Since $\nu_2(X) \le s$ and $\mathrm{Re}\,\nu_3(X) \le a$,

$$9J \le 10d_X + 71\nu_2(X) + 30\,\mathrm{Re}\,\nu_3(X) \le 10d_X + 71s + 30a.$$

Taking the integer part proves the claim. □

To apply these inequalities using fusion rules, suppose now that $K_0(\mathcal{C})$ is commutative of rank r, with simples $X_0 = 1, X_1, \ldots, X_{r-1}$. Put $d_i = \mathrm{FPdim}(X_i)$, $d = (d_i)_{i=0}^{r-1}$ and $B = \sum_i N_i N_i^{\mathsf{T}}$. The *induction matrix* has entries $M_{ai} = [I(X_i) : Z_a] = [F(Z_a) : X_i]$, where Z_a ranges over the simple objects of the center. Let L consist of the rows for the simple summands of $I(1)$. There are r such summands, each occurring once, with dimensions D/f_j [26, Theorem 2.13]. One row is $e_0^{\mathsf{T}} = (1, 0, \ldots, 0)$, corresponding to the central unit. Adjunction gives

$$M^{\mathsf{T}} M = B, \qquad L^{\mathsf{T}}\mathbf{1} = Be_0, \qquad Q := B - L^{\mathsf{T}}L, \qquad W_i := (L^{\mathsf{T}}Ld)_i.$$

Thus $W_i = W_{X_i}$ and Q is the Gram matrix of the rows outside L. Every row v^{T} of L, assigned to a formal codegree f, satisfies

$$v \in \mathbb{Z}_{\ge 0}^r, \qquad v_0 = 1, \qquad d^{\mathsf{T}} v = D/f, \qquad v_i \le \min\{B_{0i}, \lfloor\sqrt{B_{ii}}\rfloor\}.$$

The bounds follow from the first column and diagonal of $M^{\mathsf{T}}M = B$. They give a finite list of possible L. For each choice, the preceding corollaries can be tested directly. Rows assigned to equal

formal codegrees may be permuted without changing $L^{\mathsf{T}}L$, so these permutations need not be tested separately.

Example 3.7. Corollary 3.6 rules out the self-dual fusion ring R_2 with basis $\{1, X, Y, Z\}$ and products

$$\begin{aligned} X^2 &= 1 + X + Y + 2Z, & XY &= X + 2Y + 2Z, & XZ &= 2X + 2Y + 2Z, \\ Y^2 &= 1 + 2X + 2Y + 2Z, & YZ &= 2X + 2Y + 3Z, & Z^2 &= 1 + 2X + 3Y + 3Z. \end{aligned}$$

Direct multiplication checks the fusion-ring axioms. Put $\delta = 3+\sqrt{10}$, $\delta' = 3-\sqrt{10}$ and $D = 6+18\delta$. The four characters are

$$(1, \delta - 1, \delta, \delta + 1), \quad (1, \delta' - 1, \delta', \delta' + 1), \quad (1, 0, 1, -1), \quad (1, 1, -1, 0).$$

The first is the Frobenius–Perron character. Any pivotal structure is spherical, since the simples are self-dual. The rational characters vanish on a simple, so Lemma 2.3 reduces any pivotal categorification to a pseudo-unitary one. Its formal codegrees are $D, 6 + 18\delta', 3, 3$. The nonunit summands of $I(1)$ therefore have dimensions $1 + 6\delta, 2 + 6\delta, 2 + 6\delta$, and the preceding row conditions give

$$(1, r, 6 - 2r, r), \qquad (1, p, 5 - 2p, p + 1), \qquad (1, q, 5 - 2q, q + 1).$$

Since $L^{\mathsf{T}}\mathbf{1} = Be_0$, we have $r + p + q = 5$; nonnegativity gives $r \le 3$. Hence

$$W_X = r(1 + 6\delta) + (p + q)(2 + 6\delta) = 10 - r + 30\delta \ge 7 + 30\delta.$$

But $X^2 = 1 + X + Y + 2Z$ gives $d_X = \delta - 1$ and $s = a = 1$ in Corollary 3.6. Since $6 < \delta < 31/5$, $\lfloor (10\delta + 91)/9 \rfloor = 16$. The corollary would give

$$224 + 960\delta \le 32W_X \le D\big(7(\delta - 1) + 16\big) = 180 + 960\delta,$$

where $\delta^2 = 6\delta + 1$. This is a contradiction.

One additional inequality is useful when a simple object is an invertible involution.

Proposition 3.8. *If $g \ne 1$ is an invertible involution satisfying $Q_{gg} = 2$, $D/2 - W_g = 1$ and $W_g > 0$, then*

$$2W_i + DQ_{gi} \le D(d_i + 1) \qquad (i \ne 0, g).$$

Proof. Since $Q_{gg} = 2$, exactly two simple summands of $I(g)$ lie outside its low part, each with multiplicity one. The sum of all summand dimensions is D, while the second twist trace is $D\nu_2(g)$. Here $|\nu_2(g)| = 1$, so equality in the triangle inequality forces $\theta_A^2 = \nu_2(g)$ for each summand A of $I(g)$. As $W_g > 0$, one such twist is one. Thus $\nu_2(g) = 1$ and all twists are ± 1.

The first twist trace is zero, so the dimensions at each of the two twists sum to $D/2$. Outside the low part, the dimensions are therefore $D/2 - W_g = 1$ at twist 1 and $D/2$ at twist -1. The dimension-one summand restricts to g; the other has induction row $(Q_{gi} - \delta_{gi})_i$. For $i \ne 0, g$, these summands and the low part consequently contribute at least $W_i + (D/2)Q_{gi}$ to the dimension at twists ± 1 in $I(X_i)$. The polynomial $1 + \mathrm{Re}(z^2)$ is nonnegative on the circle and equals two at both 1 and -1. The identity in the proof of Proposition 3.1 therefore gives

$$2W_i + DQ_{gi} \le D(d_i + \nu_2(X_i)) \le D(d_i + 1).$$

□

Lemma 3.9. *Suppose $K_0(\mathcal{C})$ is commutative of rank $r \geq 2$, and put*

$$S = \sum_{i>0} d_i, \qquad A = \sum_{i>0} d_i N_{ii}^{i^*}, \qquad w = \sum_{j=0}^{r-1} f_j^{-2}.$$

Then

$$9(D^2 w - D) \leq D\big(3(D-1) + 2S\big), \tag{3.1}$$

$$36(D^2 w - D) \leq D\big(10(D-1) + 8S + 2A\big).$$

In particular, if $r = 4$, then

$$A \geq D - 25 - 4S + \frac{24}{D}. \tag{3.2}$$

Proof. Each low row has unit entry one and dimension D/f_j. Hence

$$\sum_{i>0} d_i W_i = \sum_j \frac{D}{f_j}\left(\frac{D}{f_j} - 1\right) = D^2 w - D,$$

using $\sum_j f_j^{-1} = 1$. Multiply the inequalities of Corollaries 3.3 and 3.5 by d_i and sum, using $\nu_2(X_i) \leq 1$, to obtain the first two claims. The second gives $A + 4S \geq 18Dw - 5D - 13$. Since $f_0 = D$, Cauchy–Schwarz gives

$$w \geq \frac{1}{D^2} + \frac{1}{r-1}\left(1 - \frac{1}{D}\right)^2.$$

For $r = 4$, substitution gives the last inequality. □

4. A uniform dimension bound

All fusion categories in this section are over $\mathbb{C}$ and have rank four, unless another rank is stated explicitly. The estimates of Section 3 yield a bound independent of any census. This bound makes the subsequent classification a finite problem; the sharper bounds in Remark 1.2 follow only after that classification.

Theorem 4.1. *Every rank-four fusion ring R admitting a pivotal categorification satisfies*

$$\mathrm{FPdim}(R) < 3600, \qquad \mu(R) \leq 59.$$

4.1. Reduction to the self-dual pseudo-unitary case.

Lemma 4.2. *Let $\mathcal{C}$ be a self-dual pivotal fusion category of rank four. Either it has a pseudo-unitary Galois conjugate, or* $\mathrm{FPdim}(\mathcal{C}) < 576$.

Proof. Write $R = K_0(\mathcal{C})$ and $D = \mathrm{FPdim}(\mathcal{C})$. Self-duality makes the pivotal structure spherical. If D is rational, it is an integer and the fusion category is pseudo-unitary by [9, Proposition 8.24]. For the rest of the proof, assume that no Galois conjugate is pseudo-unitary. Then D is irrational, and the spherical dimension character is outside the Frobenius–Perron orbit.

The field generated by spherical dimensions contains every formal codegree, including D, by [26, Corollary 2.15]. Here the *Frobenius–Perron field* means $K = \mathbb{Q}(\mathrm{FPdim}(X) : X \in \mathcal{O}(\mathcal{C}))$, the field of values of the Frobenius–Perron character; it need not equal $\mathbb{Q}(D)$ a priori. Its degree is the size of that character orbit, hence lies between two and four. Degree four leaves no other character orbit, and in degree three the other orbit is rational, so neither case is possible. If $[K : \mathbb{Q}] = 2$, then $\mathbb{Q}(D) = K$ because D is irrational. The other character field contains K, and only two dimensions remain in $R \otimes \mathbb{Q}$; it must therefore be another copy of K. Thus

$$R \otimes \mathbb{Q} \cong K \times K \tag{4.1}$$

for a real quadratic field K, and the spherical dimension character belongs to the other factor.

Denote the four formal codegrees by D, D', H, H', with primes indicating conjugation in K. We allow $H = H'$ and list one formal codegree for each character. After Galois conjugation, arrange that the spherical global dimension is $H \geq H'$. Each formal codegree is positive and greater than one. By [9, Propositions 8.21 and 8.22], $H \leq D$ and H/D is an algebraic integer. Equality would make the fusion category pseudo-unitary, so $H < D$ in the exceptional case. Put

$$n = DD', \qquad k = \frac{HH'}{DD'} \in \mathbb{Z}_{\geq 1}, \qquad p = \frac{1}{D} + \frac{1}{D'}, \qquad w = \frac{1}{D^2} + \frac{1}{(D')^2} + \frac{1}{H^2} + \frac{1}{(H')^2}.$$

Here k is the field norm of H/D, including when that element is rational. Since $D' > 1$ and $H \geq H'$, we have

$$n > D, \qquad H \geq \sqrt{kn} > \sqrt{kD}. \tag{4.2}$$

Equation (2.1) gives $D^{-1} + (D')^{-1} + H^{-1} + (H')^{-1} = 1$, and so

$$w - \frac{1}{2} = 2\left(p - \frac{1}{2}\right)^2 - \frac{2(1+1/k)}{n}. \tag{4.3}$$

Ostrik's spherical inequality (2.2), applied with global dimension H, gives

$$w \leq \frac{1}{2} + \frac{1}{2H}.$$

Set $r = D/H > 1$. By (4.2), $r < \sqrt{D/k} \leq \sqrt{D}$. Moreover, $kr > 2$. This is immediate when $k \geq 2$. Both H/D and its conjugate H'/D' are positive. When $k = 1$, H/D is therefore a totally positive algebraic-integer unit of norm one. Thus r is also such a unit, with conjugate r^{-1}; the integer $r + r^{-1} > 2$ is at least three, and $r \geq (3 + \sqrt{5})/2 > 2$.

Since $H' = krD'$, the same identity gives

$$1 = \frac{1+r}{D} + \frac{1+1/(kr)}{D'}, \qquad p = \frac{kr}{kr+1} - \frac{kr^2-1}{(kr+1)D} > \frac{2}{3} - \frac{r}{D} > \frac{2}{3} - \frac{1}{\sqrt{D}}.$$

If $D \geq 576$, then $p > 5/8$ and $H > 24$. Equation (4.3) consequently gives

$$w - \frac{1}{2} > \frac{1}{32} - \frac{4}{D} \geq \frac{7}{288} > \frac{1}{48} > \frac{1}{2H},$$

contradicting (2.2). Thus $D < 576$, as required. □

Lemma 4.3. *A pivotal fusion category of rank four with a non-self-dual simple has Frobenius–Perron dimension less than* 45*.*

Proof. A pointed fusion category has Frobenius–Perron dimension four, so assume that the fusion category is not pointed. Write the basis as $1, X, Y, Z$, where $X^* = Z$ and $Y^* = Y$.

By Lemma 2.2, we may choose a spherical structure.

There are exactly two real-valued characters of this fusion ring. Indeed, by the adjoint relation $N_{U^*} = N_U^{\mathsf{T}}$ and simultaneous unitary diagonalization, $\chi(U^*) = \overline{\chi(U)}$ for every basis element U. Extend basis duality $\mathbb{C}$-linearly. This involution has trace two on the distinguished basis; on the primitive idempotents it is a permutation, so exactly two characters are fixed, precisely the real-valued ones. If the Frobenius–Perron character is irrational, its Galois orbit has at least two members, all fixed by duality because $\mathrm{FPdim}(U) = \mathrm{FPdim}(U^*)$, and hence consists of these two characters. The spherical dimension character is real-valued, so it belongs to that orbit. Lemma 2.3 gives a pseudo-unitary Galois conjugate. If the Frobenius–Perron character is rational, the global Frobenius–Perron dimension is an integer and pseudo-unitarity follows directly from [9, Proposition 8.24].

Larson's candidate classification [17, Theorem 1.1] bounds the global Frobenius–Perron dimension of a non-self-dual pseudo-unitary fusion category of rank four by $24 + 12\sqrt{3} < 45$. The Galois conjugate has the same fusion ring, so the same bound holds for $\mathcal{C}$. □

4.2. **Concentration in the self-dual case.** Let $\mathcal{C}$ be a self-dual pseudo-unitary fusion category of rank four, with simples $1, X_1, X_2, X_3$. Put

$$d_i = \mathrm{FPdim}(X_i), \quad a_i = N_{ii}^i, \quad S = \sum_{i=1}^3 d_i, \quad A = \sum_{i=1}^3 d_i a_i, \quad w = \sum_{j=0}^3 f_j^{-2}.$$

Its fusion ring is commutative. Index the characters so that $\chi_0 = \mathrm{FPdim}$ and $f_0 = \mathrm{FPdim}(\mathcal{C})$. Character orthogonality gives

$$a_i = \sum_{j=0}^3 \frac{\chi_j(X_i)^3}{f_j}, \tag{4.4}$$

by taking the coefficient of 1 in X_i^3.

Lemma 4.4. *Suppose that* $\mathrm{FPdim}(\mathcal{C}) \geq 3600$. *Put* $M = \max_i d_i^2$ *and* $E = \mathrm{FPdim}(\mathcal{C}) - M$. *Then*

$$2 < f_j < 4 \quad (1 \leq j \leq 3), \qquad f_1 f_2 f_3 < 36, \tag{4.5}$$

$$E < \frac{7}{2}\sqrt{\mathrm{FPdim}(\mathcal{C})}. \tag{4.6}$$

There is a unique simple object X *of maximum dimension. Its fusion matrix is singular, and*

$$|\chi_j(X)|^2 < \frac{14}{\sqrt{\mathrm{FPdim}(\mathcal{C})}} \leq \frac{7}{30} < \frac{1}{4} \qquad (1 \leq j \leq 3). \tag{4.7}$$

Proof. Put $D = \mathrm{FPdim}(\mathcal{C})$, $t = \sqrt{D} \geq 60$, $q_j = f_j^{-1}$, $m = (1 - D^{-1})/3$, and $V = \sum_{j=1}^3 (q_j - m)^2$. Substitution of $w = D^{-2} + 3m^2 + V$ into (3.1) gives

$$V \leq \frac{4}{3D} + \frac{2S}{9D} - \frac{4}{3D^2} < \frac{4}{3t^2} + \frac{7}{18t} \leq \frac{37}{5400} < \frac{1}{144},$$

where $S < \sqrt{3}t < 7t/4$. Three real numbers of sum zero each have square at most two thirds of their squared norm. Therefore

$$|q_j - m| \leq \sqrt{2V/3} < \frac{1}{14}, \qquad \frac{1}{4} < \frac{11}{42} - \frac{1}{3D} < q_j < \frac{17}{42} < \frac{1}{2}.$$

This proves $2 < f_j < 4$. Writing $q_j = 1/4 + r_j$ with $r_j > 0$ gives

$$q_1 q_2 q_3 \geq \frac{1}{64} + \frac{r_1 + r_2 + r_3}{16} = \frac{1}{32} - \frac{1}{16D} > \frac{1}{36},$$

and hence the product bound.

Relabel so that $d_3^2 = M$. Since $f_j < 4$, every non-Frobenius–Perron character value on a nonunit simple has absolute value less than 2. Equations (2.1) and (4.4) imply

$$A \leq \frac{\sum_i d_i^4}{D} + 2\sum_i d_i \left(1 - \frac{d_i^2}{D}\right). \tag{4.8}$$

Initially, $\sum_i d_i^4/D \leq M$ and (3.2) give

$$E \leq 25 + 6S < 25 + 12t < 13t. \tag{4.9}$$

The coarse bound confines E to an interval on which the following comparison is effective; it is not yet the bound needed for the determinant argument. For a sharper estimate, use

$$S \leq t + \sqrt{2E}, \qquad \sum_i d_i^4 \leq (D - E)^2 + (E - 1)^2,$$

and

$$\sum_i d_i \left(1 - \frac{d_i^2}{D}\right) \leq \frac{E}{t} + \sqrt{2E}.$$

Here the largest dimension contributes $d_3E/D \leq E/t$, and the other two contribute at most $d_1 + d_2 \leq \sqrt{2E}$. Combining these estimates with (3.2) and (4.8) gives $g(E) \leq -(2E+23)/t^2 < 0$, and in particular

$$g(E) := 2E - \frac{2E^2}{t^2} - \frac{2E}{t} - 6\sqrt{2E} - 4t - 25 \leq 0. \tag{4.10}$$

Suppose $E \geq E_0 := 7t/2$. On $[E_0, 13t]$,

$$g'(E) = 2 - \frac{4E}{t^2} - \frac{2}{t} - \frac{3\sqrt{2}}{\sqrt{E}} > 2 - \frac{54}{t} - \frac{6}{\sqrt{7t}} > 0.$$

But

$$(3t-57)^2 - 252t = 9\big((t-60)(t-6)+1\big) > 0,$$

so

$$g(E_0) = 3t - \frac{113}{2} - 6\sqrt{7t} > \frac{1}{2}.$$

This contradicts (4.10) and proves (4.6). In particular $E < D/2$, so $M > D/2$ and the largest simple is unique.

Write $X = X_3$ and let N_X be its fusion matrix. Put $y_j = \chi_j(X_3)^2/f_j$ for $1 \leq j \leq 3$. Orthogonality gives $y_1 + y_2 + y_3 = E/D$. The arithmetic–geometric mean inequality now shows

$$\begin{aligned} |\det N_X|^2 &= M(f_1f_2f_3)y_1y_2y_3 \\ &< 36D\left(\frac{E}{3D}\right)^3 < \frac{343}{6\sqrt{D}} \leq \frac{343}{360} < 1. \end{aligned}$$

Thus $\det N_X = 0$, since it is an integer. Finally, $\chi_j(X_3)^2/f_j \leq E/D$ and $f_j < 4$ imply (4.7). □

4.3. **The self-dual pseudo-unitary bound.**

Proposition 4.5. *If a pseudo-unitary fusion category has rank four and all its simple objects are self-dual, then its Frobenius–Perron dimension is less than* 3600.

Proof. Put $R = K_0(\mathcal{C})$ and $D = \mathrm{FPdim}(\mathcal{C})$, and suppose $D \geq 3600$. Set $t = \sqrt{D}$, and let X be the largest simple supplied by Lemma 4.4. Write the basis as $1, X, Y, Z$, with dimensions $1, x, y, z$, and put

$$K = \mathbb{Q}(x,y,z), \qquad r = [K:\mathbb{Q}], \qquad E = D - x^2.$$

The rank of N_X is r. The rational semisimple commutative algebra $R \otimes \mathbb{Q}$ is a product of number fields. Its character orbits are the embeddings of those fields, and the Frobenius–Perron orbit corresponds to K. If a character χ lies outside this orbit, each embedding of $\mathbb{Q}(\chi(X))$ extends to its character field. Thus all conjugates of $\chi(X)$ occur in the same non-Frobenius–Perron character orbit and have absolute value less than one by (4.7). The norm of a nonzero algebraic integer has absolute value at least one, so $\chi(X) = 0$. On the Frobenius–Perron orbit the values are embeddings of the nonzero element $x \in K$, and are nonzero. Simultaneous diagonalization identifies these character values, with their multiplicities, with the eigenvalues of N_X. There are exactly r embeddings in the Frobenius–Perron orbit, so exactly r eigenvalues are nonzero. Hence

$$\operatorname{rank} N_X = r. \tag{4.11}$$

The principal block of N_X on $1, X$ is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & a \end{smallmatrix}\right)$, so its rank is at least two. Its determinant vanishes by Lemma 4.4. Thus $r = 2$ or 3.

Rank two is impossible. Write

$$N_X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & a & b & c \\ 0 & b & e & h \\ 0 & c & h & k \end{pmatrix}.$$

The upper left block is invertible and its Schur complement is exactly the lower right block. The Schur-complement rank formula [14, §0.8.5(b)] therefore gives $e = h = k = 0$. Consequently $XY = bX$ and $XZ = cX$, so $y = b$ and $z = c$ are positive integers. The second character in the Frobenius–Perron orbit has formal codegree

$$1 + (x')^2 + b^2 + c^2 < 4.$$

It follows that $b = c = 1$. Thus Y and Z are distinct self-dual invertible objects [8, Corollary 3.3.10]. Their product is a fourth invertible basis element, necessarily X: it cannot be $1, Y$, or Z by cancellation. This gives $D = 4$, a contradiction.

Rank three is impossible. Now $R \otimes \mathbb{Q} \cong \mathbb{Q} \times K$ with $[K : \mathbb{Q}] = 3$. The remaining character ρ is integer-valued and has formal codegree strictly between two and four, hence equal to three. Since $\rho(X) = 0$, its values on Y, Z are each ± 1. Orthogonality with the positive character excludes equal signs. Relabeling gives

$$\rho = (1, 0, 1, -1), \qquad z = y + 1. \tag{4.12}$$

Applying ρ to products and using Frobenius reciprocity gives nonnegative integers b, c, s such that

$$\begin{aligned} XY &= bX + cY + cZ, & Y^2 &= 1 + cX + sY + sZ, \\ YZ &= cX + sY + (s+1)Z. \end{aligned} \tag{4.13}$$

The trace of N_Y is $b + 2s + 1$. Subtracting $\rho(Y) = 1$ gives

$$\mathrm{Tr}_{K/\mathbb{Q}}(y) = b + 2s. \tag{4.14}$$

By (4.6),

$$E = 1 + y^2 + (y+1)^2 < \frac{7}{2}t, \qquad x^2 = t^2 - E > (t-4)^2.$$

The dimension equation for Y^2 gives $cx < y^2 < E/2$, whence

$$c < \frac{7t}{4(t-4)} < 2.$$

Thus $c \le 1$. The equation for XY gives

$$\delta := y - b = \frac{c(2y+1)}{x} \ge 0, \qquad \delta^2 \le \frac{2E-3}{D-E} < \frac{7}{t - 7/2} < 1.$$

For either nonidentity embedding of K, write $u = y'$ and $v = x'$. By (4.7), $|v| < 1/2$. Applying the embedding to the equation for Y^2 gives

$$u^2 = 1 + cv + s(2u+1).$$

If $u \ge -1/2$, then $c \le 1$ and $s \ge 0$ give $u^2 > 1/2$, hence $u > 1/\sqrt{2}$. Its formal codegree would satisfy

$$f' = 1 + v^2 + u^2 + (u+1)^2 > 3 + \sqrt{2} > 4,$$

contrary to (4.5). Thus $y', y'' < -1/2$, and (4.14) gives

$$2s = (y - b) + y' + y'' < 1 - 1 = 0,$$

a contradiction. This completes the proof. □

Proof of Theorem 4.1. Let $\mathcal{C}$ be a pivotal categorification of R and put $D = \mathrm{FPdim}(\mathcal{C})$. If it has a non-self-dual simple, Lemma 4.3 gives $D < 45$. Otherwise, Lemma 4.2 either gives $D < 576$ or supplies a pseudo-unitary Galois conjugate with the same fusion ring. Proposition 4.5 then gives $D < 3600$. The preliminary bound $\mu(R)^2 \le D$ implies $\mu(R) < 60$, hence $\mu(R) \le 59$ since fusion coefficients are integers. □

5. Proof of Theorem 1.1

The fifteen Grothendieck rings arising from Table 1 are pairwise nonisomorphic: the global Frobenius–Perron dimension is an invariant of fusion rings, and the fusion categories $\mathrm{Vec}_{C_2\times C_2}$ and Vec_{C_4}, as well as $PSU(2)_6$ and the principal even part of S', do not have the same fusion rules. Thanks to Lemma 2.2, it is therefore enough to show that there are, up to isomorphism, at most fifteen fusion rings of rank four admitting a spherical categorification.

By Theorem 4.1, such a ring must have global Frobenius–Perron dimension less than 3600. We show below that there are 8977 fusion rings of rank four whose global Frobenius–Perron dimension is less than 3600. To prove Theorem 1.1, we therefore need to rule out all but fifteen of these fusion rings. This is done using the exclusion criteria stated in Sections 2 and 3.

The enumeration uses the following elementary criterion for associativity.

Lemma 5.1. *Let A be a commutative ring and let M be a A-module with a A-bilinear multiplication and a two-sided identity 1. For $x \in M$, write L_x for the operator mapping $y \in M$ to xy. The multiplication is associative and commutative if and only if $L_xL_y = L_yL_x$ for all $x, y \in M$.*

Proof. Let $x, y \in M$. For an associative and commutative multiplication, $L_xL_y = L_{xy} = L_{yx} = L_yL_x$. Conversely, applying $L_xL_y = L_yL_x$ to 1 gives $xy = yx$. Hence, for all $z, t, u \in M$, we have

$$(tu)z = z(tu) = L_zL_t(u) = L_tL_z(u) = t(zu) = t(uz)$$

and the multiplication is associative as well. □

Recall that every fusion ring of rank four is commutative. Let R be a free $\mathbb{Z}$-module of rank four with basis $\{b_0 = 1, b_1, b_2, b_3\}$ and multiplication $b_ib_j = \sum_k N_{ij}^k b_k$, where $N_{ij}^k \geq 0$ are integers, satisfying the commutativity, unit, duality, and Frobenius reciprocity axioms. Up to isomorphism, we can assume that either $b_i^* = b_i$ for all i, or $b_1^* = b_1$ and $b_2^* = b_3$.

For every $u \in \{0, 1, 2, 3\}$, let N_u be the multiplication matrix of b_u. The matrix of the operator L_{b_u}, in the usual column convention, is N_u^T. Consequently, thanks to Lemma 5.1, R is a fusion ring if and only if the matrices N_1, N_2, and N_3 pairwise commute.

In the self-dual case, the multiplication matrices are of the form

$$N_1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & a & b & c \\ 0 & b & d & e \\ 0 & c & e & f \end{pmatrix}, \quad N_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & b & d & e \\ 1 & d & g & h \\ 0 & e & h & i \end{pmatrix}, \quad N_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & c & e & f \\ 0 & e & h & i \\ 1 & f & i & j \end{pmatrix}$$

and their pairwise commutativity is equivalent to

$$\begin{aligned} ad - b^2 + bg + ch - d^2 - e^2 + 1 &= 0, \\ ae - bc + bh + ci - de - ef &= 0, \\ be - cd + dh - eg + ei - fh &= 0, \\ af + bi - c^2 + cj - e^2 - f^2 + 1 &= 0, \\ bf - ce + di - eh + ej - fi &= 0, \\ df - e^2 + gi - h^2 + hj - i^2 + 1 &= 0. \end{aligned} \tag{5.1}$$

In the non-self-dual case, they are of the form

$$N_1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & a & b & b \\ 0 & b & c & d \\ 0 & b & d & c \end{pmatrix}, \quad N_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & b & c & d \\ 0 & d & e & f \\ 1 & c & e & e \end{pmatrix}, \quad N_3 = N_2^{\mathsf{T}}$$

and their pairwise commutativity is equivalent to

$$\begin{aligned} ac - b^2 + 2be - c^2 - d^2 + 1 &= 0, \\ ad - b^2 + be + bf - 2cd &= 0, \\ b(c-d) + d(e-f) &= 0, \\ -c^2 + d^2 - e^2 + f^2 - 1 &= 0. \end{aligned} \tag{5.2}$$

The following commands compute all solutions of (5.1) or (5.2) with $\mathrm{FPdim}(R) < 3600$ (and so $\mu(R) \leq 59$ by Lemma 2.1); they are 8977 up to fusion ring isomorphism:

```
python3 supplementary/census/generate.py
python3 supplementary/census/verify_census.py
```

Run all commands from the package root, alongside `manuscript.tex` and the `supplementary` directory. They require Python 3.10 or later and SymPy; census generation also requires a C++17 compiler. The supplementary guide gives installation instructions. No external census is used.

Theorem 2.4 excludes 1676 rings containing a rank-two fusion subring with $x^2 = 1 + ax$ and $a \geq 2$. The command

```
python3 supplementary/stages.py rank-two
```

checks these exclusions and leaves 7301 rings. Among these rings, the command

```
python3 supplementary/stages.py d-number
```

rules out 6541 fusion rings that have a formal codegree failing the d-number condition of Theorem 2.5. It remains 760 fusion rings.

Let R be one of these 760 fusion rings and assume that it is the Grothendieck ring of some spherical fusion category $\mathcal{C}$. Each character of R gives candidate values for $\dim_{\mathcal{C}}(V)$ for every simple object $V \in \mathcal{O}(\mathcal{C})$. These numbers must satisfy some conditions according to Subsections 2.2 and 2.4: they are nonzero real numbers, $\dim_{\mathcal{C}}(V) = \dim_{\mathcal{C}}(V^*)$ for all simple objects V, Ostrik's inequality (2.2), and, for every formal codegree f of R, the field $\mathbb{Q}(\dim(\mathcal{C})/f)$ is an abelian extension of $\mathbb{Q}$ and $\dim(\mathcal{C})/f$ has integral norm. The command

```
python3 supplementary/stages.py spherical
```

eliminates 595 rings for which no character satisfies all of the above conditions, leaving 165 rings.

Among these 165 rings, the published family restrictions in Subsection 2.4 give 90 further exclusions. The command

```
python3 supplementary/stages.py families
```

checks the full fusion rules of those families and leaves 75 rings.

Now, if R is one of these 75 fusion rings, we write S_R for the set of characters of R passing all the arithmetic conditions above. This is a non-empty set by construction. Among these 75 fusion rings R, we consider those for which $\dim(\mathcal{C})$ and $\mathrm{FPdim}(\mathcal{C})$ are Galois conjugates for all characters in S_R. For each character in S_R, Lemma 2.3 tells us that $\mathcal{C}$ can be assumed pseudo-unitary, possibly after replacing $\mathcal{C}$ by a Galois conjugate depending on the character. The command

```
python3 supplementary/stages.py central
```

excludes 60 rings in this subset: every possible low induction matrix violates at least one necessary condition in Section 3. There remain 15 fusion rings, identified with Table 1 by the command

```
python3 supplementary/stages.py models
```

This proves Theorem 1.1. □

The non-necessarily pivotal case is the subject of our work in progress, *Classifying Grothendieck rings of general fusion categories of rank four*.

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